

Sherard Dalaguit

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EDUCATION

University of California, Irvine

Expected 12/2026

B.S. Computer Science | B.S. Software Engineering

GPA: 3.75 | Dean's Honor List

- **Relevant Coursework:** Data Structures & Algorithms, Computing in Python, Software Engineering Principles, Artificial Intelligence, Software Architecture, Software Development Life Cycle, UX/UI Design, Data Modeling

EXPERIENCE

Panasonic Avionics Corporation

01/2026 – Present

Software Engineer

Irvine, CA | Capstone

- Built a Python backend validation service to prevent incompatible aircraft software deployments by enforcing hardware, software, and configuration constraints prior to fleet rollout.
- Implemented a deterministic resolution engine evaluating proposed deployments in <5s by traversing dependency graphs across 1,500+ aircraft targets.
- Architected a Neo4j dependency graph schema supporting semantic version ranges, hard vs. soft requirements, and incompatibilities across 250,000+ deployable artifacts.
- Reduced simulated fleet-scale deployment failures by 90%+ by partnering with a Principal Architect to shift validation to pre-deployment constraint evaluation with actionable incompatibility explanations.

OpenLyst

05/2025 – 09/2025

Founding Engineer (Self-Initiated Startup)

Remote

- Launched a production remote job aggregation system processing 2,000+ new listings weekly via GitHub Actions pipelines that scrape multiple sources and purge stale listings, normalizing data into a unified schema.
- Engineered resilient ingestion pipelines with deduplication and failure handling, reducing duplicate listings by ~40% and ensuring consistent data quality.
- Designed a MongoDB-backed data model optimized for high-frequency writes and fast querying, enabling near real-time job updates and personalized delivery.
- Automated serverless email delivery workflows with Next.js API routes and SendGrid, delivering personalized job digests to 375+ subscribers with >98% delivery success.

PROJECTS

TagalogAI: Voice-Based AI Tagalog Coach

Next.js, MongoDB, OpenAI, Vapi, ElevenLabs

- Developed a low-latency AI conversation partner for Tagalog speaking practice, integrating OpenAI and Vapi APIs and maintaining sub-1.5s response times for uninterrupted natural dialogue.
- Architected a modular conversation evaluation engine analyzing grammar, vocabulary, and fluency, delivering structured feedback with 90%+ detection accuracy in controlled testing.
- Introduced progressive coaching modes atop a shared feedback pipeline, allowing new learning flows to be added without modifying core system logic.

Rizzlytics: AI Dating App Analyzer

TypeScript, React, Next.js, MongoDB, OpenAI

- Created a multi-modal AI system for analyzing dating photos, profiles, and message threads, producing structured feedback from heterogeneous inputs.
- Designed mode-aware analysis pipelines integrating user uploads, external media storage, and OpenAI Vision and language models with schema-validated outputs.
- Improved production robustness by adding retry, fallback, and error-handling mechanisms around external asset access and AI inference.

SWEOverflow: StackOverflow Clone App

TypeScript, React, Next.js, MongoDB

- Built a full-stack, StackOverflow-style knowledge exchange platform, implementing authentication, voting, reputation, and search features.
- Integrated AI-generated answers through the OpenAI API, speeding up user response times by >75%.

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C, C++, HTML, CSS, SQL
- **Technologies:** React, Next.js, Node.js, TailwindCSS, MongoDB, Mongoose, Neo4j, OpenAI, Vapi, Git, Figma